EXECUTIVE SUMMARY · MACHINE LEARNING I · UNIVERSIDAD EAN

Prioritizing public contracts at risk of cost overruns (SECOP II)

Team SECOPE: Sebastián Gómez, Kimberly Moreno, Luly Castiblanco · October 2026 · kimmoreno03.github.io/ml1-trabajo-final

The problem

One in four competitively awarded public contracts in Colombia ends up costing more than its awarded value. Between 2021 and 2024, value additions in the 244,518 completed contracts we analyzed totaled COP 19.7 trillion, 16% to 20% of the awarded amount each year. Oversight teams cannot review every contract, so they need to know where to look first.

What we built

Using open SECOP II data, we trained a model that estimates, at the moment of award, the probability that a contract will receive value additions. It only uses information known at award time, such as the type of good or service, budget, agreed duration, competition and the contracting entity's sector and region. We trained on 2021–2023 contracts, validated by grouping public entities so the model is always tested on entities it has not seen, and tested once on 2024. A gradient boosting model was selected with a rule fixed in advance.

Results

Reviewing the 10% of 2024 contracts the model ranked riskiest, 65 out of 100 had a cost overrun, compared with 25 when contracts are picked at random. That 10% held 65% of all the money added that year. The model finds more overruns than reviewing the largest contracts first (25% vs. 18% of all cases), while the largest contracts capture more money, so the two criteria work best together. Results held across another time window, an alternative definition of overrun and within each procurement method.

Recommendation

Use the published ranking of 31,486 ongoing 2025–2026 contracts to decide what to review first, working down the list as capacity allows. At 10% capacity, that means 3,149 contracts. Add a small random sample to monitor performance and bias, and retrain every year. The model prioritizes; it does not prove wrongdoing. Every review needs human judgment.

Limits

It does not cover direct contracting, which has no awarded value, and it predicts whether an overrun happens, not its size.